NAME James Stuart

ChE 210 Exam #1 1 October 2025

To receive full credit, you must show all of your work.

(1) (50 pts.) Phosphoric acid (PA) can be produced by reacting calcium phosphate (CP) with a stoichiometric quantity of sulfuric acid (SA). Calcium sulfate (CS) is also a byproduct of this reaction. Two streams enter the reactor: (**Stream 1**) has a species mass flow rate of 10,000 kgCP/hr of CP which is suspended in an unknown amount of water (W) and (**Stream 2**) a 95 wt% aqueous solution of SA. There are two exit streams: (**Stream 3**) a 40 wt% aqueous solution of PA and (**Stream 4**) a solid of composed of CS and water which has a mole ratio of 2 moles of water for each mole of CS.

Species	Molar Mass
CP ($Ca_3(PO_4)_2$)	310.18
SA (H_2SO_4)	98.08
PA (H_3PO_4)	97.994
CS ($CaSO_4$)	136.14
W (H_2O)	18.02

(a) Write a balanced chemical reaction equation for this reaction. (5 pts.)

$$Ca_3(PO_4)_2 + 3H_2SO_4 \longrightarrow 2H_3PO_4 + 3CaSO_4$$

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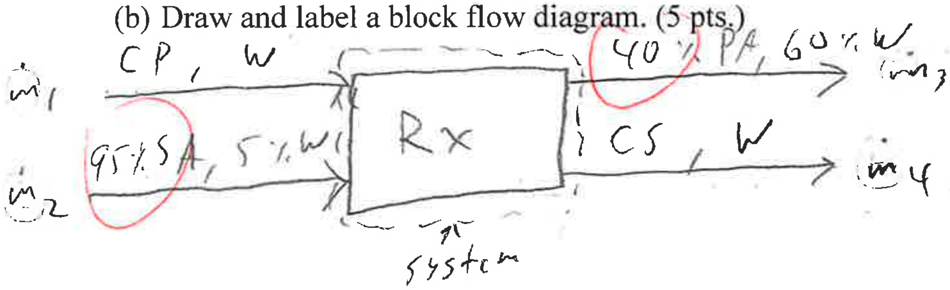
(b) Draw and label a block flow diagram. (5 pts.)



$w_{CP,1}\dot m_1 = 10{,}000 \frac{kg\,CP}{hr}$

$\frac{mol\,W}{mol\,CS} = 2$

? $\frac{18.02g}{136.14g} \times 2$?

mass ratio: $\frac{wt\,W}{wt\,CS} = 0.2647$

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(c) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. In the table below, give values for 4a, 5a, 5b, 5c, and 5d along with a brief explanation for the value you give. Notice that the value for 4b is given with an explanation. Compute the DOF. (10 pts.)

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	Number	Explanation
4a. Stream Variables	8	8 components
4b. System Variable	0	No reactants in product streams. The reaction is complete.
Total Independent Variables	~~8~~ 9	8 + 0 = 8
5a. Specified Flows	~~0~~ 1	10,000
5b. Specified Stream Compositions	3	1 in stream 2, 1 in 3 1 in 4
5c. Specified System Performances	0	
5d. Material Balance Equations	5	5 species cross system boundary
Total Independent Equations	~~8~~ 9	
DOF	0	fully specified 8 − 8 = 0

[Problem 2 continued]

(c) (30 pts.) Calculate the species molar flow rates of the ammonia leaving the system, that is, A_2 and A_4. This calculation is likely to require solving a quadratic equation.

$X_{2A} = 2X_{4A}$ ✓

$3.718 \frac{\text{kmol}}{\text{hr}} = A_2 + A_4$ ✓

$3.718 = X_{2A}\dot{n}_2 + X_{4A}\dot{n}_4$ ✓

$3.718 = 2X_{4A}\dot{n}_2 + X_{4A}\dot{n}_4$ ✓

$3.718 = X_{4A}(2\dot{n}_2 + \dot{n}_4)$

$3.718 = ((1-X_{4w})\dot{n}_4(2\dot{n}_2 + \dot{n}_4)$

$3.718 = (\dot{n}_4 - X_{4w}\dot{n}_4)(2\dot{n}_2 + \dot{n}_4)$

$3.718 = (\dot{n}_4)^2 + 2\dot{n}_2\dot{n}_4 - X_{4w}2\dot{n}_2\dot{n}_4 - \dot{n}_4\dot{n}_2$

$\dot{n}_4\dot{n}_2 + \dot{n}_4^2 - 2X_{4w}\dot{n}_2\dot{n}_4 - X_{4w}\dot{n}_4^2$

$(\dot{n}_4)^2(1-X_{4w}) - \dot{n}_4\dot{n}_2(1-2X_{4w})$

$X_{4A} = (1-X_{4w})\dot{n}_4$?

$\frac{A_4}{55.56 + A_4} = X_{A4}$ ✓

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You need more balance eqns.
$\dot{n}_1 + \dot{n}_3 = \dot{n}_2 + \dot{n}_4$ for example, etc.

$N_2 =$ 33.459 kmolN/hr $A_2 =$ — kmolA/hr $W_4 =$ 55.56 kmolW/hr $A_4 =$ — kmolA/hr

(d) (5 pts.) What is the percent recovery of the ammonia, that is, what percentage of the ammonia entering the separator is removed with the water stream?

$\%\text{ recovery} = \frac{X_{A4}\dot{n}_4}{3.718 \frac{\text{kmol}}{\text{hr}}} \times 100$

2Y 2

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PROBLEM 2

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$$N: N_1 = N_2$$

$$A: 0.1\dot{m}_1 = X_{A2}\dot{m}_2 + X_{A4}\dot{m}_4$$

$$W: \dot{m}_3 = X_{W4}\dot{m}_4$$

$$\dot{m}_2 + \dot{m}_4 = 2000 \frac{kg}{hr}$$

$$X_{2A} = 2X_{4A}$$

$$\dot{m}_1 = 1000 \frac{kg}{hr}$$

$$X_{A1} = 0.1$$

$$\dot{m}_3 = 1000 \frac{kg}{hr}$$

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(2) **(50 pts.)** A gas mixture of nitrogen N_2 (N) and ammonia NH_3 (A) enters the bottom of a separator as **Stream 1** with mass flow rate $\dot{m}_1 =$ 1000 kg/hr and a mole fraction of the ammonia $x_{A1} = 0.10$. **Stream 2** exits the top of the separator containing a gas mixture of nitrogen and ammonia. Pure liquid water H_2O (W) enters the top of the same separator as **Stream 3** at a mass flow rate of $\dot{m}_3 =$ 1000 kg/hr and a liquid mixture of water and ammonia exits the bottom as **Stream 4**. The nitrogen and water do not mix during the process. The mole fractions of the ammonia in the two exiting streams are known to be related by $x_{2A} = 2x_{4A}$. Assume steady state. Molecular Weights: M_A = 17 kg/kmol for ammonia, M_N =28 kg/kmol for nitrogen, and M_W = 18 kg/kmol for water.

(a) (5 pts.) Calculate the <u>mass</u> fraction of A in Stream 1, that is w_{1A}.

$$\frac{0.1 \text{ mol } A}{0.9 \text{ mol } N} = \frac{1.7 \text{ kg } A}{25.2 \text{ kg } N} \rightarrow \frac{1.7 \text{ kg}}{1.7 \text{ kg} + 25.2 \text{ kg}} = \boxed{0.0632 = w_{1A}}$$

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$w_{1A} =$ 0.0632 kgA/kg

(b) (10 pts.) Calculate the species <u>molar</u> flow rates for the inlet ammonia A_1, nitrogen N_1, and water W_3. $\rightarrow \dot{n}_3$

$N: \; N_1 = x_{N2}\dot{n}_2$

$A: \; 0.1\dot{n}_1 = x_{A2}\dot{n}_2 + x_{A4}\dot{n}_4$

$W: \; \dot{n}_3 = x_{W4}\dot{n}_4$

$\dot{n}_1 + \dot{n}_3 = \dot{n}_2 + \dot{n}_4$

$A: \; 3.718 \frac{\text{kmol}}{\text{hr}} = x_{A2}\dot{n}_2 + 2x_{A2}\dot{n}_4$

$= x_{A2}(\dot{n}_2 + 2\dot{n}_4)$

$w_{A1} = 0.0632\,\dot{m}_1 = 63.2 \frac{\text{kg}}{\text{hr}}$

$x_{A1}\dot{n}_1 = 3.718 \frac{\text{kmol}}{\text{hr}}$

$x_{N1}\dot{n}_1 = 9x_{A1}\dot{n}_1 = 33.459$

$\dot{n}_2 = \frac{N_1}{x_{N2}}$

$$W_3 = \frac{1000 \text{ kg W}/\text{hr}}{18 \text{ kg}/\text{kmol}}$$

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$A_1 =$ 3.718 kmolA/hr $\quad N_1 =$ 33.459 kmolN/hr $\quad W_3 =$ 55.56 kmolW/hr

[Problem 1 continued]

(d) Calculate the species molar flow rates for the four non-water components in the four streams in kmol/hr. That is CP_1, SA_2, PA_3, and CS_4 where the subscripts are stream numbers. Also, write a balance on water and calculate W_1, the water entering the process with CP_1. Show all your calculations and write your final answers in the spaces provided below. (30 pts.)

$CP_1 = 10000 \frac{kg}{hr} \times \frac{1\ kmol}{310.18\ kg} = 32.24 \frac{kmol}{hr}$

CP: $CP_1 = CP_{cons}$

SA: $0.95\dot{m}_2 = SA_{cons}$

PA: $0.4\dot{m}_3 = PA_{gen}$

CS: $CS_4 = CS_{gen}$

$SA_2 = SA_{cons} = 3CP_{cons} = 3CP_1 = 96.72 \frac{kmol}{hr}$

$PA_3 = PA_{gen} = 2CP_{cons} = 64.48 \frac{kmol}{hr}$

$CS_4 = CS_{gen} = 3CP_{cons} = 96.72 \frac{kmol}{hr}$

$3CP_{cons} = SA_{cons}$

$2CP_{cons} = PA_{gen}$

$3CP_{cons} = CS_{gen}$

W: $W_1 + W_2 = W_3 + W_4 \rightarrow W_1 = W_3 + W_4 - W_2$

$W_1 = 691.71 \frac{kmol}{hr}$

$W_4 = 2CS_4 = 193.44 \frac{kmol}{hr}$

$\frac{mass\ SA}{mass\ W} = 19 \rightarrow \frac{5\ g\ W}{95\ g\ SA} = \frac{0.27747\ mol\ W}{0.9695\ mol\ SA} = \frac{0.2862\ mol\ W}{mol\ SA\ 2}$

$W_2 = 27.68 \frac{kmol\ W}{hr}$

$\frac{mass\ PA}{mass\ W} = \frac{2}{3} \rightarrow \frac{60\ g\ W}{40\ g\ PA} = \frac{3.3296\ mol\ W}{0.4082\ mol\ PA} \rightarrow \frac{8.1568\ mol\ W}{mol\ PA}$

$W_3 = 525.95 \frac{kmol\ W}{hr}$

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1 50
2 30
T 80

Species molar flow rate. Subscripts indicate stream numbers.

CP_1 (kmol/hr)	SA_2 (kmol/hr)	PA_3 (kmol/hr)	CS_4 (kmol/hr))	W_1 (kmol/hr)
32.24	96.72	64.48	96.72	691.71